

POKHARA UNIVERSITY

Level: Bachelor

Programme: BE

Course: Numerical Methods (Old)

Semester: Fall

Year : 2023

Full Marks: 100

Pass Marks: 45

Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Find where the graph of $y = x-3$ and $y = \ln(x)$ intersect using bisection method. Get the intersection value correct to four decimal places. 8
- b) Find value of $\sqrt{18}$ using Newton Raphson method. 7
2. a) Find the root of the equation $3x + \sin x - e^x = 0$ corrected up to 3 decimal places using Secant Method. 7
- b) Given populations of a city in different year as shown in table below. 8

x (in year)	1891	1901	1911	1921	1931
f(x)(population of that year in 0000)	46	66	81	93	101

Observe the data carefully and answer the following questions:

- i. Which method is applicable for finding population in the year 1860 and why? Also find the population of that year.
 - ii. If you are asked to find the change in population between the year 1925 to 1930, how will you find? (mention formula and steps)
3. a) Fit a power function model of the form $y = ax^b$ from the given data points. 8

x	1	2	3	4	5
y	0.5	2	4.5	8	12.5

- b) Use Romberg method to evaluate: 7
- $$\int_0^{3\pi/2} e^x \sin x \, dx$$
4. a) Solve the following system of equations using partial pivoting method. 8
- $$x + y + z = 4, \quad x + 4y + 3z = 8, \quad x + 6y + 2z = 6$$

- b) Obtain the solution of following system using Gauss-Seidel iteration method. 7

$$2x_1 + x_2 + x_3 = 5$$

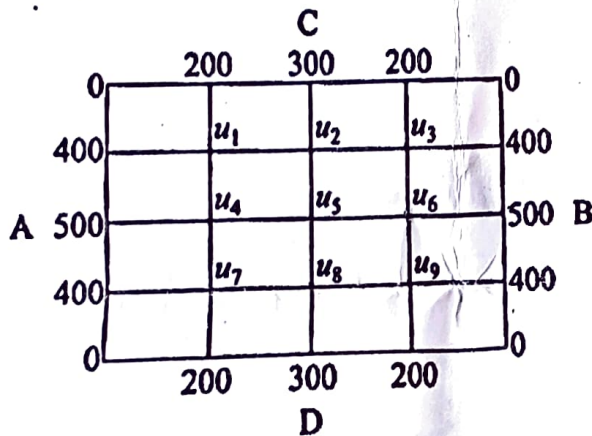
$$3x_1 + 5x_2 + 2x_3 = 15$$

$$2x_1 + x_2 + 4x_3 = 8$$

5. a) Determine the largest eigenvalue and the corresponding eigenvector of the matrix: 7

$$A = \begin{bmatrix} 1 & -3 & 2 \\ 4 & 4 & -1 \\ 6 & 3 & 5 \end{bmatrix} \text{ using the power method.}$$

- b) Solve the Laplace equation $U_{xx} + U_{yy} = 0$ for the square mesh with boundary values as shown in the figure. 8



6. a) Solve the following equation for $y(0.2)$ using shooting method: 9
 $\frac{d^2y}{dx^2} + 2\frac{dy}{dx} - 3y = 6x$ Given $y(0) = 0, y'(0) = 1$.
- b) Use Picard's method to approximate the value of y when $x=0.1, x=0.2$ and $x=0.4$, given that $y=1$ at $x=0$ and $dy/dx = 1+xy$ correct to three decimal places (Use upto second approximations). 6
7. Write short notes on: (Any two) 2x5
- Newton's backward differentiation formulas
 - Partial differential equation and their examples
 - Algorithm of fixed point iteration method